

BunkVauth Technical Report

Engineering Project: Zero-Trust Continuous Biometric Authentication

1. Dataset Overview

The BunkVauth system utilizes a behavioral biometric dataset captured via keystroke and mouse dynamics. The dataset consists of 30 unique users with 50 samples each, creating a robust baseline for identification.

Behavioral Features (Input Vectors)

Feature Name	Description
avg_dwell_time	Mean duration (ms) for which keys are held down.
std_dwell_time	Standard deviation of key hold duration (consistency).
avg_flight_time	Mean interval (ms) between key release and next press.
std_flight_time	Rhythm stability of the user.
typing_speed	Keystrokes per second (KPS).
avg_mouse_speed	Pixels moved per millisecond (Mouse agility).

2. Machine Learning Model

Algorithm: Random Forest Classifier
Configuration: 100 Estimators (Trees), Gini impurity criterion.

Random Forest was selected for its high performance in multi-class classification and its ability to identify the most significant behavioral traits through feature importance.

3. Evaluation Metrics

- Accuracy (61.33%):** The percentage of sessions correctly assigned to the authorized user.
- Precision (0.6305):** Reliability of the positive classification (Low False Positives).
- Recall (0.6259):** Ability to find all instances of the real user (Low False Negatives).
- ROC AUC (0.9769):** The models superior capability to distinguish between any two random users.

Conclusion

The high ROC AUC score validates that behavioral biometrics is a viable second factor for SSO security, providing a non-intrusive layer of continuous protection.